

Track C — How to Run Guide

This guide provides step-by-step instructions for running Track C (RGTP pruning evaluation) on top of a trained Track B checkpoint.

Prerequisites

1. Activate the Virtual Environment

Run commands from the repository root with the venv active:

```
cd D:\Thesis\Ego4d-LiteSTA
.\local_extraction\.venv\Scripts\Activate.ps1
```

2. Verify Imports

Quick import check:

```
python -c "from local_extraction.trackC import trackC_pruning; print('OK')"
```

```
python -c "from local_extraction.trackC import trackC_compare_metrics;
print('OK')"
```

3. Required Data

Ensure you have:

- **Extracted frames:** `local_extraction/v2/extracted_frames/<uid>/<frame:07d>.jpg`
- **Stage B val manifest:** from `local_extraction/runs/Track_A/trackA_stageB_*/head_val.jsonl`
- **Track B checkpoint:** `local_extraction/runs/Track_B/checkpoints/trackB_best.pt` (or any `trackB_final_*.pt`)

Optional:

- **Pre-extracted tokens:** set `data.tokens_root` in `local_extraction/configs/trackC.yaml` if you want faster inference.

Quick Start

Default Run (uses `trackC.yaml`)

```
python local_extraction/trackC/trackC_pruning.py
```

Defaults:

- Config: `local_extraction/configs/trackC.yaml`
- Checkpoint: auto-detects latest/best Track B checkpoint
- Val manifest: auto-detects latest Stage B run
- Pruning: enabled with `rgtp.rate` from config

Baseline (No Pruning)

```
python local_extraction/trackC/trackC_pruning.py --no_pruning
```

Override RGTP Rate

```
python local_extraction/trackC/trackC_pruning.py --rgtp_rate 0.30
```

Use a Specific Checkpoint

```
python local_extraction/trackC/trackC_pruning.py --checkpoint  
"local_extraction/runs/Track_B/checkpoints/trackB_best_mAP_0.3904_20251220_024  
940.pt"
```

Switch Backbone (VideoMAE)

```
python local_extraction/trackC/trackC_pruning.py --video_backbone videomae_ego
```

Config Overrides (YAML)

Edit `local_extraction/configs/trackC.yaml` to pin specific inputs:

- `evaluation.checkpoint`
- `evaluation.stageB_run`
- `evaluation.val_manifest`

- `evaluation.candidate_limit` , `evaluation.normalize_ttc`
- `rgtp.rate` , `rgtp.min_keep` , `rgtp.temporal_decay`

Rate sweeps and smoke tests are also configured there:

- `rate_sweep.enabled` , `rate_sweep.rates`
- `smoke_test.enabled` , `smoke_test.max_samples` , `smoke_test.test_rates`

Outputs

All outputs go to `local_extraction/runs/Track_C/` :

```
local_extraction/runs/Track_C/
├── metrics/
│   ├── trackC_val_rateXX_<timestamp>.json
│   └── trackC_val_rateXX_<timestamp>_summary.json
└── plots/ (optional)
```

Utilities

Compare Track B vs Track C Metrics

```
python local_extraction/trackC/trackC_compare_metrics.py
```

Plot Track C Metrics Over Time

```
python local_extraction/trackC/trackC_plots.py --runs_dir
local_extraction/runs/Track_C --out_dir local_extraction/runs/Track_C/plots
```

CLI Reference

```
usage: trackC_pruning.py [-h] [--config CONFIG] [--video_backbone
{resnet18,videomae_ego}]
                        [--rgtp_rate RGTP_RATE] [--checkpoint CHECKPOINT] [--
no_pruning]
                        [--hotspot {on,off}] [--clip {on,off}]

options:
```

```
--config CONFIG          Config name to load (default: trackC)
--video_backbone {resnet18,videomae_ego}
                          Override backbone selection
--rgtp_rate RGTP_RATE    Override pruning rate (0.0-0.95)
--checkpoint CHECKPOINT  Override checkpoint path
--no_pruning             Disable pruning (baseline)
--hotspot {on,off}       Enable/disable hotspot priors (override config)
--clip {on,off}          Enable/disable CLIP re-ranking (override config)
```

For full parameter explanations, see [local_extraction/trackC/CONFIGURABLE_OPTIONS.md](#) .